Risk and Return

Chapter 7

Introduction to Financial Management (Pope)

Rates of Return

The rate of return (holding period return / percentage return) is a function of the periodic cash flows and any price changes

For stocks:

$$\% \, r = \frac{\boldsymbol{dividend + capital\ gain}}{\boldsymbol{initial\ price}}$$

$$\frac{dividend}{initial\ price} + \frac{capital\ gain}{initial\ price}$$

Dividend Yield ***Capital Gains Yield***

Return on an Investment

Periodic return

- Return over a specific period
- Can be averaged to understand over time

Holding period return

- Return over an entire range of periods
- Simple percentage change from start to end of investment

Geometric average return

- Return implied by compounding over an entire range of periods
- Found using TVM equations

Example 1:

Suppose you purchase a share of Microsoft stock in mid-October of 2022 for $242.12. One year later, Microsoft has risen to $326.67. In October of 2024, you decide to sell your share for the market price of $418.16. What is your holding period return? What is the average annual return over those 3 years? What is the geometric average annual return?

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$$r_{2023} = \frac{326.67 - 242.21}{242.21} = 0.349 = 34.9\%$$

$$r_{2024} = \frac{418.16 - 326.67}{326.67} = 0.280 = 28\%$$

$$r_{avg} = \frac{0.349 + 0.28}{2} = 0.315 = 31.5\%$$

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$$r_{geo} = \left(\frac{FV}{PV}\right)^{\frac{1}{t}} - 1 = (1 + r_{HP})^{\frac{1}{t}} - 1 = \left(\frac{418.16}{242.12}\right)^{\frac{1}{2}} - 1 = 0.314 = 31.4\%$$

Return in the Market

Market indices provide a measure of investment performance (return) for the overall market

Dow Jones Industrial Average (*the Dow*)

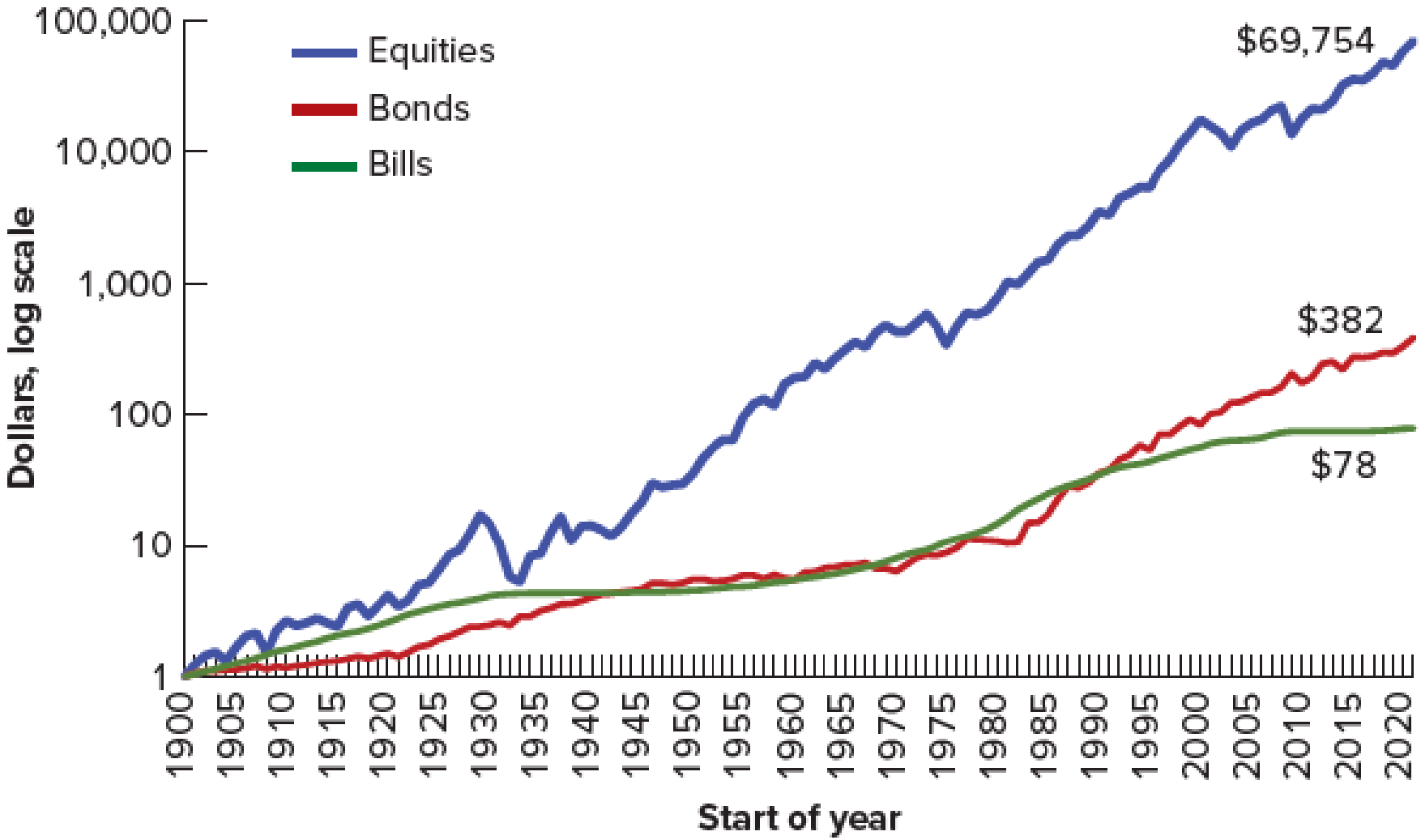
- one share of 30 large, “well-known” firms
- Microsoft, Apple, IBM, Walmart, AMEX, Home Depot, etc.

Standard & Poor’s Composite (*the S&P 500*)

- capitalization weighted index, 500 firms make up 80% of total market value
- Best indicator of average performance

Past Returns

If you invested $1 in 1900, the graph depicts the value of that investment over time.



How to Think About Risk?

Risk can be defined as a *measure of the uncertainty in a set of potential outcomes for an event i.e., how those outcomes differ from what is expected.*

- “Risky” implies not only the potential for loss but also for uncertain gain (risk does not take sides)
- Risk is an *ex-ante* measure
- Measure the level of risk through expected value and standard deviation.

Expected Value

Expected Value is the probability-weighted average of an outcome (X)

$$Expected\ value = \sum_{i=1}^{N} p_i X_i = \bar{X}$$

where X_i is outcome i (return) and p_i is the respective probability of that outcome (return) occurring

What does it mean though?

Example 3:

Suppose you and I play a simple game. When I flip a coin, if the coin lands on heads, I will pay you $10. If the coin land on tails, you will pay me $10. What is your expected outcome (payoff) for this game?

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$$Expected\ value = \sum_{i=1}^{N} p_i X_i = \bar{X}$$

$$\mathrm{E}(V) = (50\% \times \$10) + (50\% \times -\$10) = (0.5 \times 10) + (0.5 \times -10) = 0$$

Variance and Standard Deviation

Variance is the probability-weighted average of squared deviations from the expected value (mean) an outcome (X)

$$variance\ (X) = \sum_{i=1}^{N} p_i * (X_i - \bar{X})^2 = \sigma^2$$

where X_i is outcome i (return), p_i is the respective probability of that outcome (return) occurring and $\bar{X}$ is the expected outcome (return)

$$standard\ deviation = \sqrt{\sigma^2} = \sigma$$

Example 4:

Consider our game from before. When I flip a coin, if the coin lands on heads, I will pay you $10. If the coin land on tails, you will pay me $10. What is the variance and standard deviation for this game?

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$$variance(X) = \sum_{i=1}^{N} p_i * (X_i - \bar{X})^2 = \sigma^2$$

$$var(X) = 0.5 \times (10 - 0)^2 + 0.5 \times (-10 - 0)^2 = 100$$

$$sd(X) = \sqrt{\sigma^2} = \sqrt{100} = 10$$

Example 5:

Now consider two games. Game A: when I flip a coin, if the coin lands on heads, I will pay you $10; if the coin land on tails, you will pay me $10. Game B: when I flip a coin, if the coin lands on heads, I will pay you $10,000; if the coin land on tails, you will pay me $10,000. Which game should you play?

Example 5:

Game A: when I flip a coin, if the coin lands on heads, I will pay you $10; if the coin land on tails, you will pay me $10. Game B: when I flip a coin, if the coin lands on heads, I will pay you $10,000; if the coin land on tails, you will pay me $10,000.

$$\mathrm{E}(V_A) = (0.5 \times 10) + (0.5 \times -10) = 0$$

$$\mathrm{E}(V_B) = (0.5 \times 10{,}000) + (0.5 \times -10{,}000) = 0$$

$$var(X_A) = 0.5 \times (10 - 0)^2 + 0.5 \times (-10 - 0)^2 = 100$$

$$var(X_B) = 0.5 \times (10{,}000 - 0)^2 + 0.5 \times (-10{,}000 - 0)^2 = 100{,}000{,}000$$

$$sd(X_A) = \sqrt{100} = 10$$

$$sd(X_B) = \sqrt{100{,}000{,}000} = 10{,}000$$

Technical Note on Variance

Historical/real data is often considered uniform in distribution, implying each observation has the same probability (1/n), so,

$$population\ var(X) = \frac{1}{N} * \sum_{i=1}^{N} (X_i - \bar{X})^2 = \sigma_{pop}^2$$

The above formula is for a population variance (you have all the observations) but most of the time we only have a sample, so,

$$sample\ var(X) = \frac{1}{N-1} * \sum_{i=1}^{N} (X_i - \bar{X})^2 = \sigma_{sample}^2$$

Verizon (VZ)

The table present the annual returns of Verizon Communications over the a 5 year period.

Year	Annual Return	Deviations from Average Return	Squared Deviation
2017	3.98%	-0.00302	0.00001
2018	11.26%	0.06980	0.00487
2019	13.84%	0.09558	0.00914
2020	-0.13%	-0.04409	0.00194
2021	-7.55%	-0.11827	0.01399
Total	21.39%	0	0.02995

$$\mathrm{E}(r_{VZ}) = \sum_{i=1}^{5} p_i X_i = \frac{21.39\%}{5} = 4.28\%$$

$$var(r_{VZ}) = \frac{1}{N-1}\sum_{i=1}^{N}(X_i - \bar{X})^2 = \frac{0.02295}{5-1} = 0.0075$$

$$sd(r_{VZ}) = \sqrt{0.0075} = 0.0865 = 8.65\%$$

Ticker	Company	SD (%)	Ticker	Company	SD (%)
X	U.S. Steel	76.4	UNP	Union Pacific	21.4
PCG	Pacific Gas & Electric	55.4	DIS	Disney	21.2
MRO	Marathon Oil	48.9	GOOG	Alphabet	20.1
NEM	Newmont Mining	36.7	SBUX	Starbucks	18.7
AMZN	Amazon	28.3	WMT	Walmart	17.7
BA	Boeing	27.4	XOM	ExxonMobil	19.4
CPB	Campbell Soup	22.6	PFE	Pfizer	16.6
IBM	I B M	22.5	MCD	McDonald's	14.4
F	Ford	22.1	KO	Coca-Cola	12.6
INTC	Intel	22.1	**S&P 500**		**12.0**